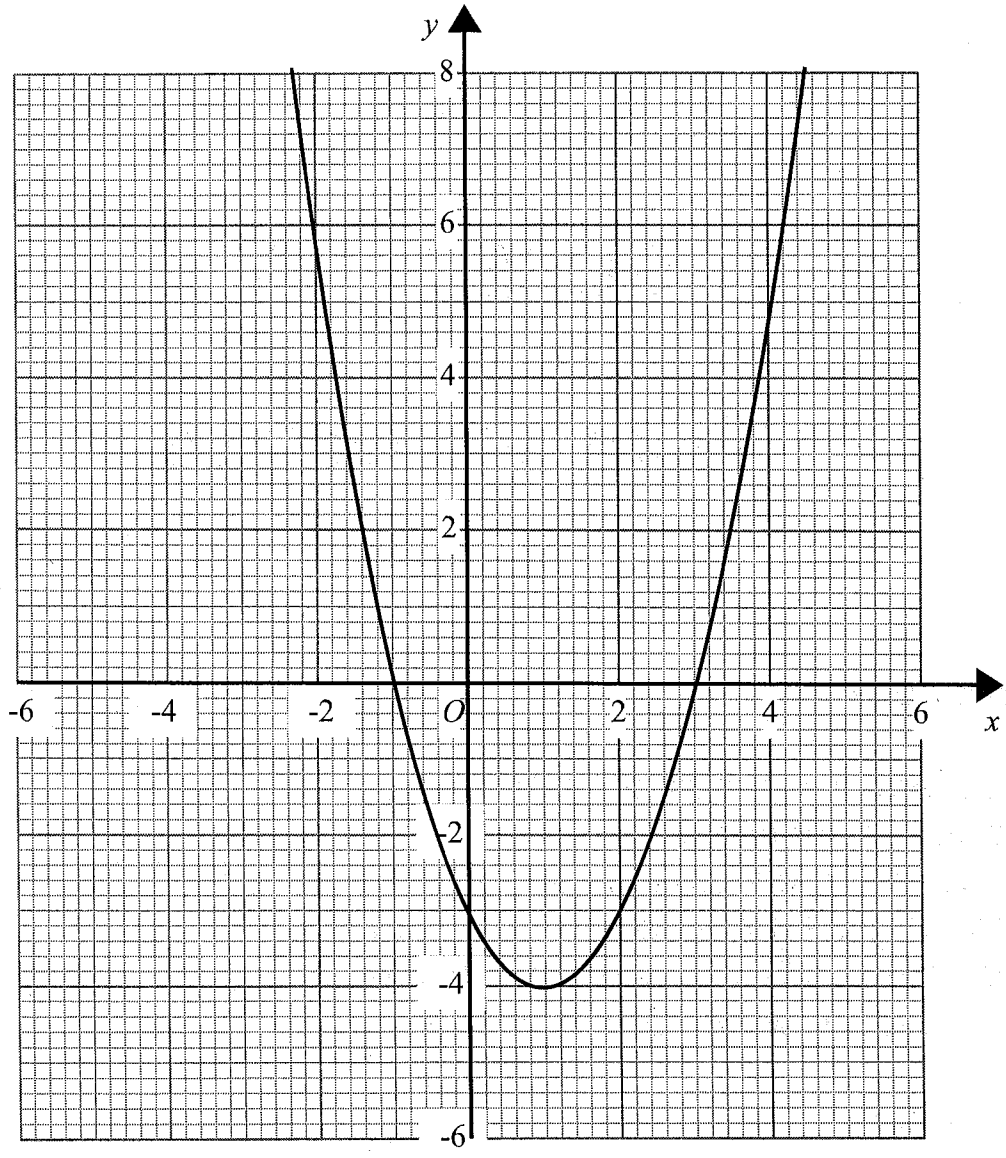
1 Here is the graph of $y = x^2 - 2x - 3$



(a) Write down the turning point of the graph $y = x^2 - 2x - 3$

(1 , −4)

(1)

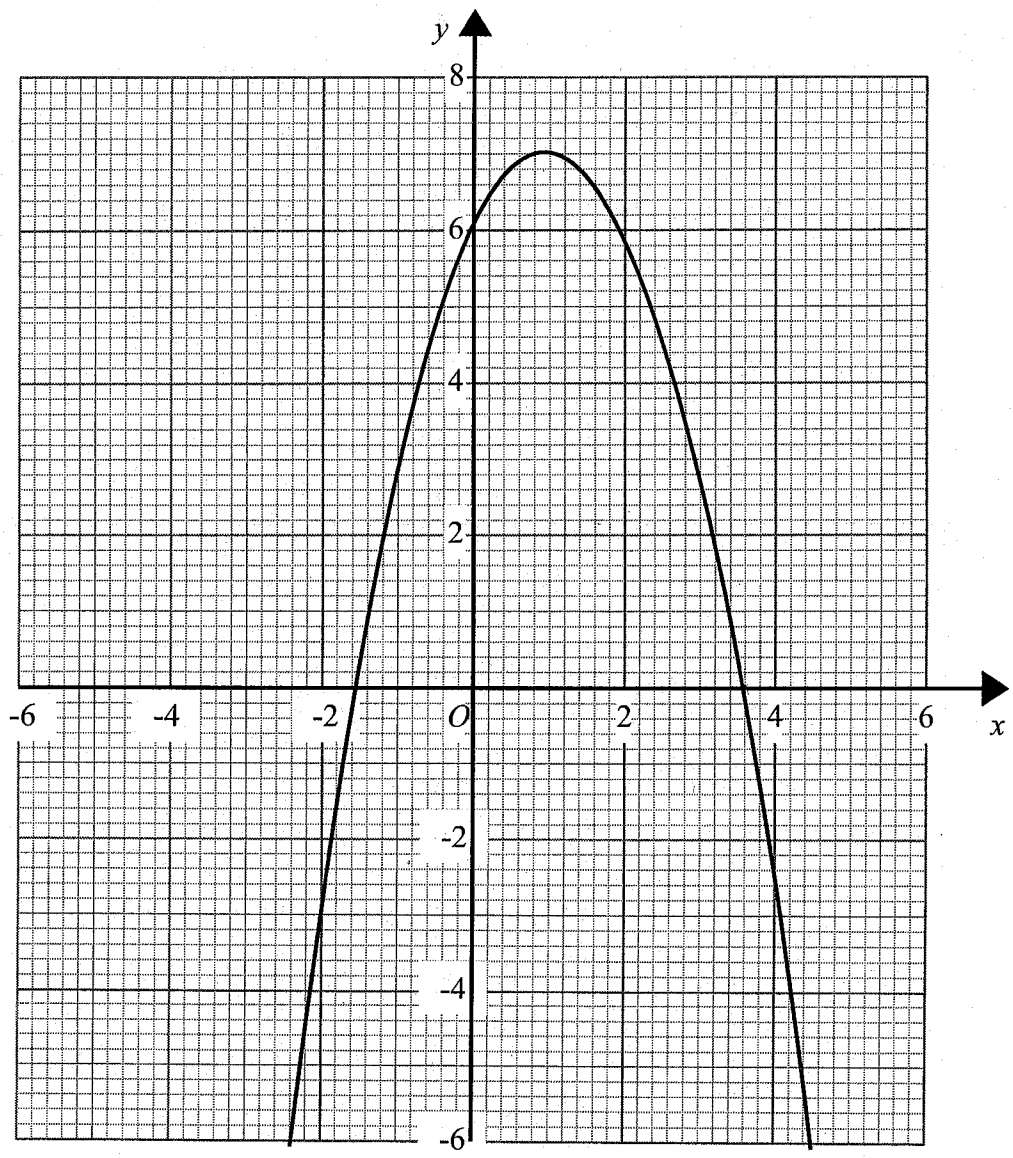
(b) Use the graph to find the roots of the equation $x^2 - 2x - 3 = 0$

−1 and 3

(2)

(Total for question 1 is 3 marks)

2 Here is the graph of $y = 2x + 6 - x^2$



(a) Write down the turning point of the graph $y = 2x + 6 - x^2$

(1 , 7)

(1)

(b) Use the graph to find the roots of the equation $x^2 = 2x + 6$

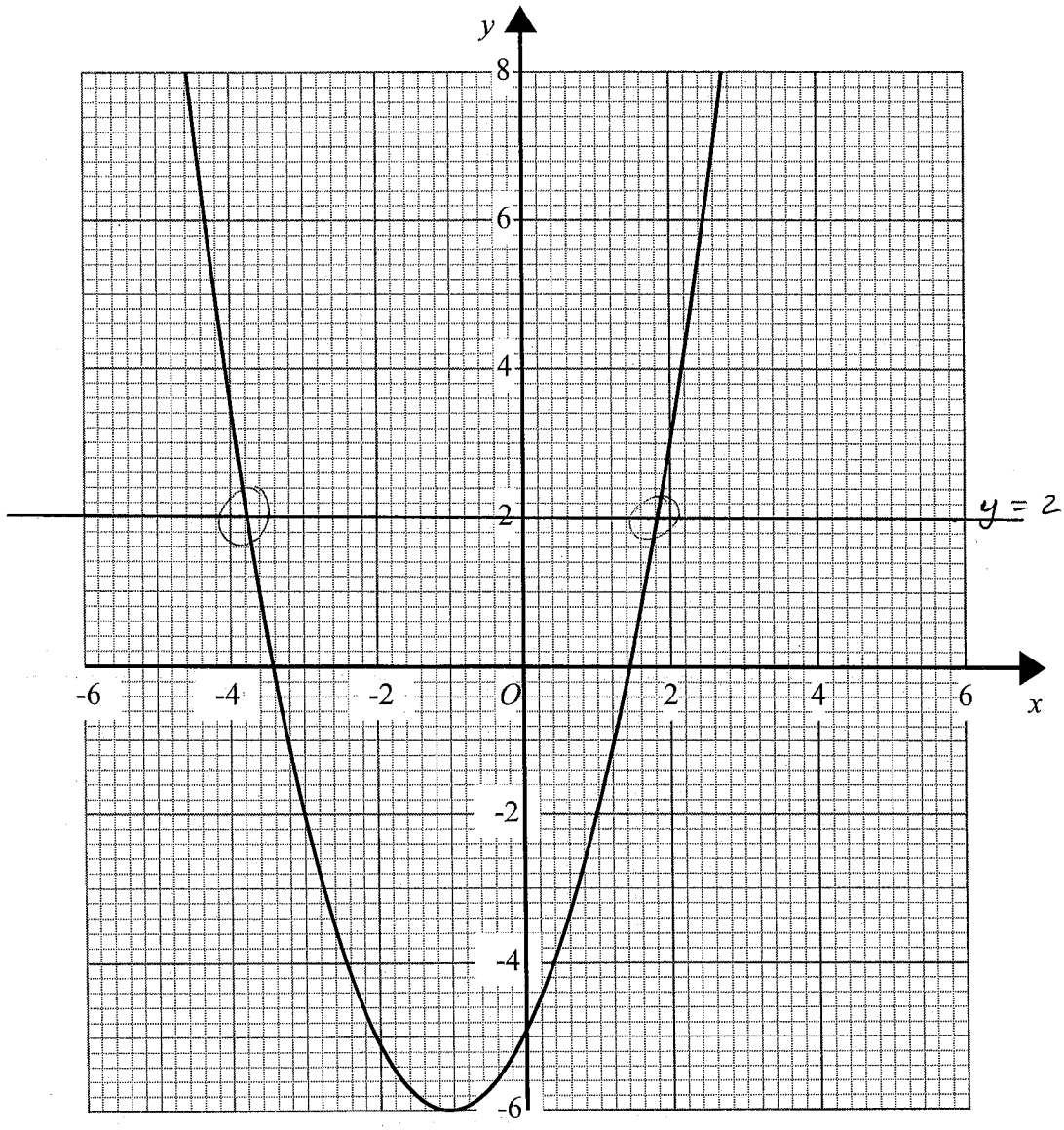
−1.6 and 3.6

(2)

(Total for question 2 is 3 marks)

accept −1.6 to −1.5
3.5 to 3.6

3 Here is the graph of $y = x^2 + 2x - 5$



(a) Write down the turning point of the graph $y = x^2 + 2x - 5$

(−1 , −6)

(1)

(b) Use the graph to find the roots of the equation $x^2 + 2x - 5 = 2$

−3.8 and 1.8

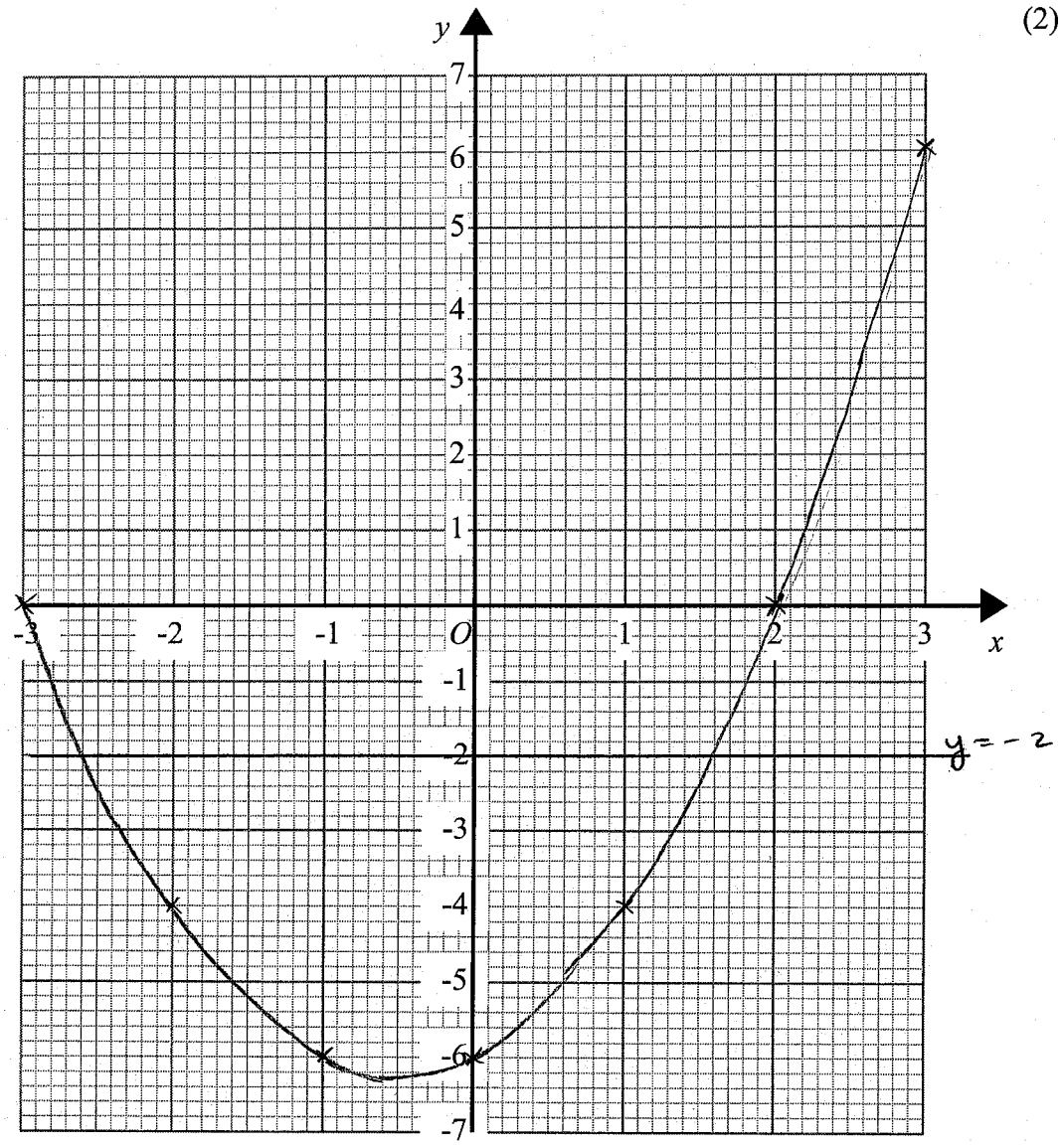
(2)

(Total for question 3 is 3 marks)

4 Complete the table of values for $y = x^2 + x - 6$

x	-3	-2	-1	0	1	2	3
y	0	-4	-6	-6	-4	0	6

(2)



(a) On the grid draw the graph of $y = x^2 + x - 6$ for values of x from –3 to 3 (2)

(b) Use the graph to find estimates of the solutions to the equation $x^2 + x - 6 = -2$

-2.6 and 1.6

(2)

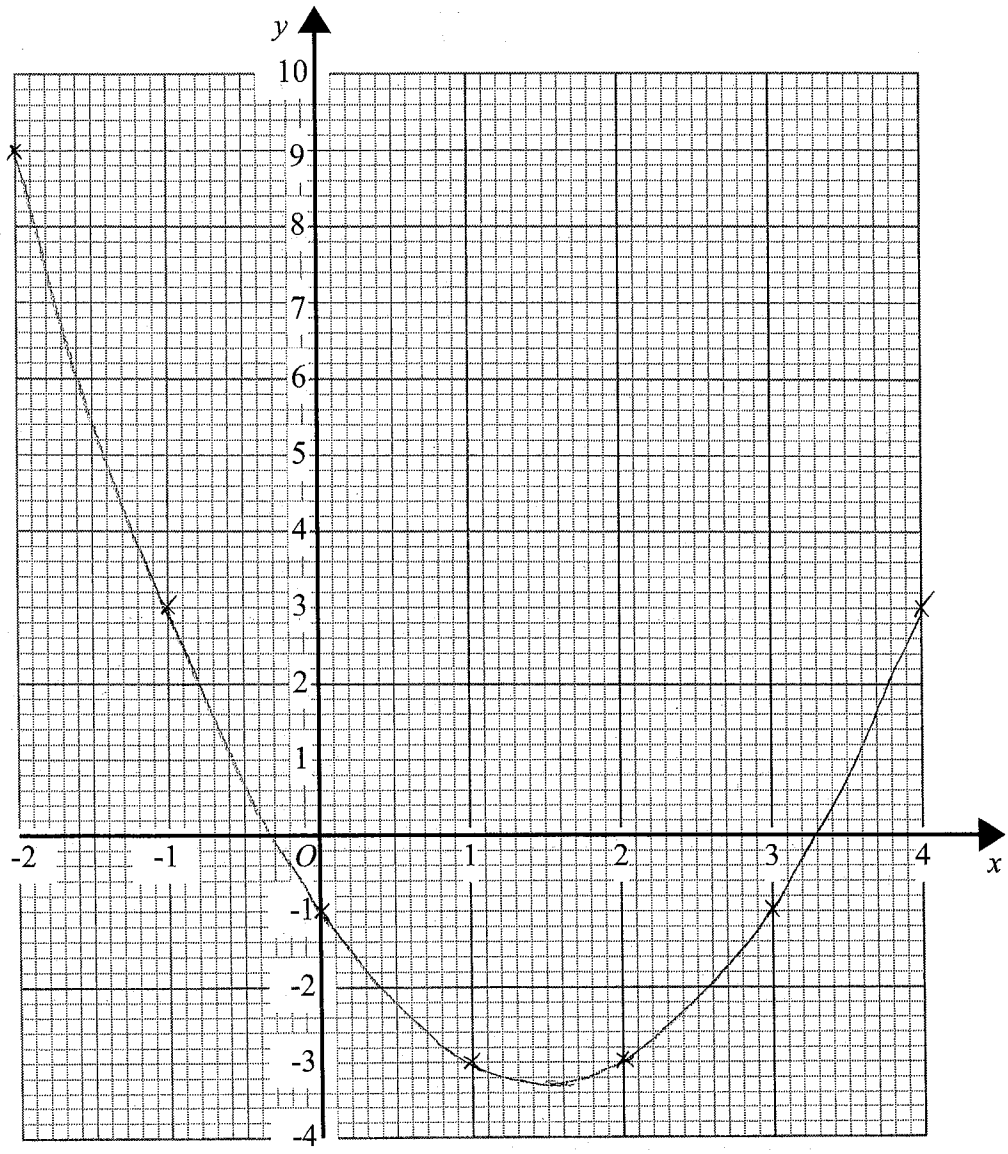
(Total for question 4 is 6 marks)

-2.6 to -2.5
1.5 to 1.6

5 Complete the table of values for $y = x^2 - 3x - 1$

x	-2	-1	0	1	2	3	4
y	9	3	-1	-3	-3	-1	3

(2)



(a) On the grid draw the graph of $y = x^2 - 3x - 1$ for values of x from –2 to 4 (2)

(b) Use the graph to find an estimate of the turning point of the graph $y = x^2 - 3x - 1$

(1.5, -3.25)

(2)

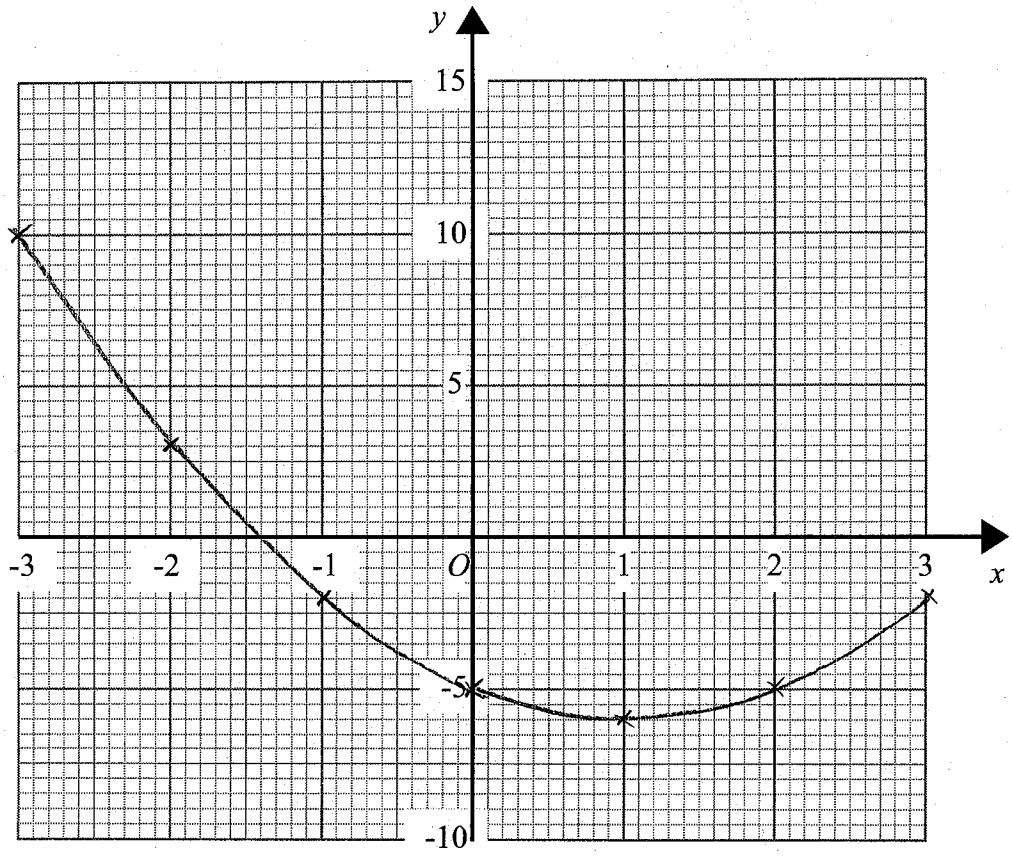
(Total for question 5 is 6 marks)

x: 1.5 only y: -3.1 to -3.5

6 Complete the table of values for $y = x^2 - 2x - 5$

x	-3	-2	-1	0	1	2	3
y	10	3	-2	-5	-6	-5	-2

(2)



(a) On the grid draw the graph of $y = x^2 - 2x - 5$ for values of x from –3 to 3 (2)

(b) Use the graph to find an estimate~~s~~ of ~~the~~ a solution~~s~~ to the equation $x^2 = 2x + 5$

-1.4

.....

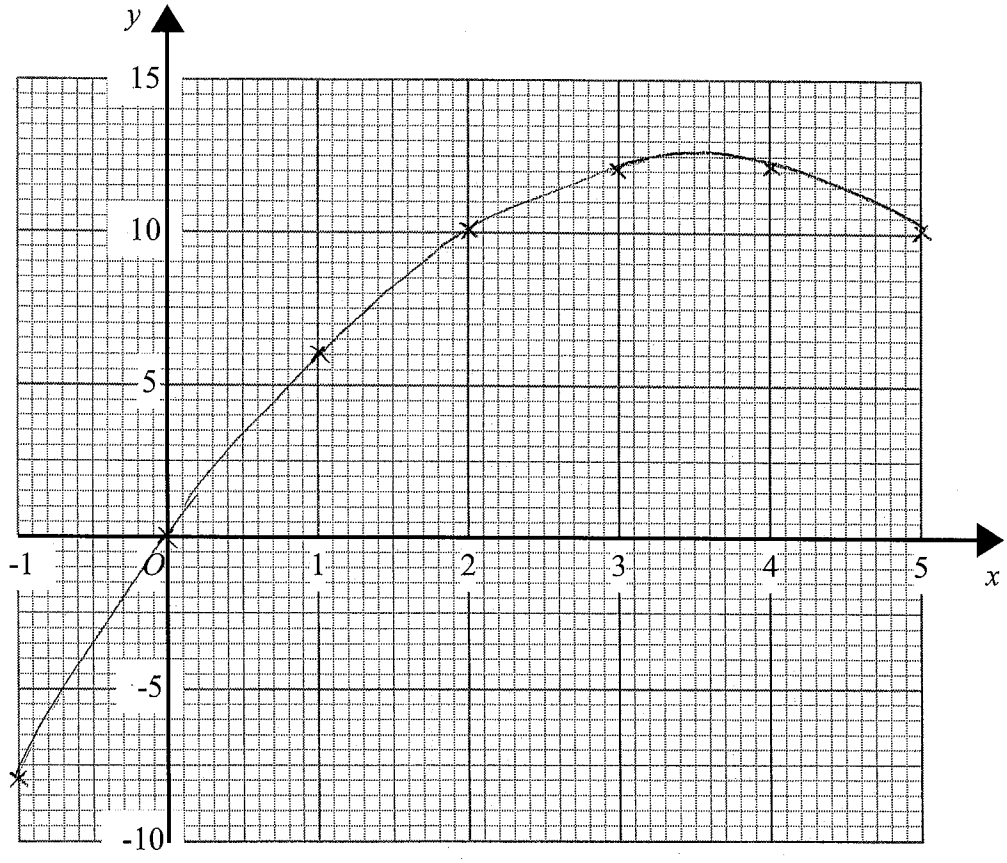
-1.5 to -1.3 ~~(2)~~ 1

(Total for question 6 is 6 marks)

7 Complete the table of values for $y = 7x - x^2$

x	-1	0	1	2	3	4	5
y	-8	0	6	10	12	12	10

(2)



(a) On the grid draw the graph of $y = 7x - x^2$ for values of x from –1 to 5 (2)

(b) Use the graph to find an estimate of the turning point of the graph $y = 7x - x^2$

3.5, 12.5

3.5 only, 12 – 13 (2)

(c) Find the solutions to the equation $7x - x^2 = 0$

$x(7 - x) = 0$

$x = 0 \quad x = 7$

$x = 0 \quad x = 7$

(2)

(Total for question 7 is 8 marks)